

Standard Operating Procedure

Mechanical Seal Replacement on Rotating Equipment — SOP-MNT-118 Rev. 6

SYNTHETIC DOCUMENT - created for a Smart India Hackathon prototype demonstration. Not real plant data. No sponsor information is used.

1. Scope

This procedure covers replacement of cartridge mechanical seals on centrifugal pumps in hydrocarbon service. It does not cover double seals with barrier fluid systems, which are covered by SOP-MNT-121.

2. Permits Required

Permit	Issued by	Validity
Hot work permit (if welding)	Area authority	One shift
Confined space entry (if applicable)	Safety officer	One shift
Equipment isolation certificate	Operations	Until work complete
Line breaking permit	Area authority	One shift

3. Procedure

1. Obtain equipment isolation certificate and confirm electrical isolation at the motor control centre.
2. Depressurise and drain the pump casing to the closed drain system.
3. Purge with nitrogen until hydrocarbon content is below 1 percent LEL.
4. Record flush line differential pressure before dismantling. This value is required for failure analysis.
5. Remove coupling guard and disconnect coupling.
6. Withdraw the cartridge seal as an assembly. Do not dismantle in the field.
7. Inspect the seal faces and photograph any scoring or deposits.
8. Fit the replacement cartridge, torquing gland bolts evenly to 45 Nm.
9. Re-establish flush and confirm 3 bar differential before releasing for start-up.
10. Complete the work order and attach the failure photographs.

4. Records

A completed work order, the recorded flush differential pressure, and photographs of the failed seal faces shall be attached to the equipment history file within two working days of completion.